Improving Course Allocation Efficiency via Auction-Based Enrollment: A Simultaneous Sealed-Bid Approach

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University course enrollment systems often suffer from inefficiencies, including misallocation of limited seats, strategic behavior under priority-based systems, and limited ability for students to express preference intensity. In this paper, we propose and study an auction-based mechanism for course allocation which models enrollment as a resource allocation problem with heterogeneous student valuations. We previously proposed the use of combinatorial auctions for capturing complex preferences, but in this paper, we focus on a simpler and more practical use of a simultaneous second-price, sealed-bid (SPSB) framework where students submit independent bids for individual courses subject to budget constraints.

We analyze the efficiency, incentive properties, and implementability of this mechanism, comparing it to both existing enrollment systems as well as select alternatives such as lottery-based and random serial dictatorship approaches. Our model examines how SPSB auctions can improve allocative efficiency while maintaining tractability and usability in real-world academic settings.

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1 Introduction

Course enrollment at large universities presents a fundamental resource allocation problem: a limited number of seats must be assigned to students with heterogeneous and often competing preferences. At the University of Virginia (UVA), this process is currently managed through a priority-based enrollment system by assigning registration times based on decrementing academic year with random serial dictatorship ordering within an academic year. While straightforward for students, this system exhibits several inefficiencies that often lead to suboptimal outcomes for students.

One primary limitation of the current UVA enrollment system is its inability to capture preference intensity. Students are constrained to expressing binary preferences (either enrolling in a course or not) without any mechanism to indicate how strongly they value one course relative to another. As a result, a student who is relatively indifferent over a given elective course may receive a seat over another student who critically needs the same course in order to graduate on time simply because of this random time-slot allocation. This lack of expressiveness can lead to ineffective allocations of scarce seats, particularly in high-demand courses and core classes (Figure 1), causing the latter to become a sort of bottleneck towards timely graduation and convenient course scheduling. Considering single seat courses, possible outcomes are restricted to (A,B) and (B,A). Assignment (B,A) is Pareto Optimal, however via random serial dictatorship assignment this outcome is only reached 50% of the time.

	Course A	Course B
Course A	(6,8)	(6,4)
Course B	(6,8)	(6,4)

Fig. 1. Inefficient Single Seat Course Allocation with Current UVA Implementation

Additionally, the time-based nature of UVA's enrollment system introduces inequities and strategic behavior. As previously mentioned, the only variable factoring into the assignment of enrollment times is class year, save for a few edge case exceptions[1]. While such systems are often designed to ensure perceived fairness, they inherently advantage students with earlier time slots and disadvantage those who must select from a diminished set of remaining options. This frequently results in these disadvantaged students experiencing a last-minute scramble for available courses where students must adjust schedules under uncertainty about which courses will remain open, often leading to inefficient or undesirable schedules.

Furthermore, the current system incentivizes strategic behavior that can distort true demand. Students often enroll in backup or placeholder courses due to uncertainty about waitlist outcomes, further contributing to artificially inflated demand and complicating the enrollment process.

These shortcomings highlight a broader issue: in the existing enrollment mechanism, there exists a need for a method to aggregate student preferences and allocating resources accordingly. Auction-based mechanisms provide a natural framework for addressing this problem by allowing students to express preference intensity through bids, thereby enabling allocations that more closely align with students' individual valuations.

In this paper, we propose and study an auction-based course allocation mechanism tailored to UVA. Our initial approach sought to explore combinatorial auctions to capture complex preferences

[1]Namely Echols Scholars, who supersede the random serial dictatorship and receive the earliest enrollment time; additional exceptions are made for student athletes and ROTC students, given the time intensive nature of their additional responsibilities.

across course bundles, but such a mechanism would be computationally intensive and difficult for participants to understand. Instead, we focus on a simpler and more practical simultaneous sealed-bid framework, in which students submit independent bids for individual courses subject to budget constraints. This approach aims to balance efficiency, tractability, and usability to provide a realistic alternative to UVA's current enrollment system.

2 Prior Work

Auction-based mechanisms for course enrollment have been explored both in practice and in the academic literature as a means of improving efficiency relative to traditional enrollment systems such as lotteries or priority-based assignment. These mechanisms are motivated by the observation that universities, acting as allocators of limited course seats, typically lack information about student preferences, thus auctions provide a natural framework for eliciting this information and allocating seats to those who value them most highly.

2.1 Existing Institutional Implementations

A number of schools and universities have already implemented auction-based course registration systems. One of the first examples is the course allocation mechanism at the University of Chicago Booth School of Business [Flores and Keedy, 2011] which employs a multi-round auction with a clearing price rule and dynamic point endowments. Students receive an initial allocation of bidding points and earn additional points over time, allowing them to strategically allocate resources across courses and semesters. A similar mechanism is used at Northwestern's Kellogg School of Management [Kellogg School of Management, nd] where a multi-round auction with refundable bids and clearing prices approximates a second-price structure, thus encouraging truthful bidding behavior.

In contrast, simpler auction formats have been adopted at schools such as Colorado College [Benson et al., 2013]. Students are given a fixed number of non-transferable points and participate in a single-round, first-price sealed-bid (FPSB) auction. Courses are allocated to the highest bidders, with waitlists formed from unsuccessful bids. While this system is easy to implement and understand, it has been shown to induce strategic behavior, as students shade bids based on the expected behavior of other students rather than true valuations.

More complex hybrid systems have also been experimented with. For example, earlier versions of the Wharton School's course allocation mechanism included multi-round bidding as well as a secondary market in which students could buy and sell course seats. While this increased flexibility, it introduced opportunities for arbitrage, collusion, and inequitable outcomes. These concerns ultimately led to the adoption of alternative mechanisms [Financial Times, 2014].

2.2 Academic Literature on Course Allocation Mechanisms

The academic literature on course allocation mechanisms builds on foundational work in mechanism design and auction theory. Early contributions formalize course registration as a resource allocation problem under capacity constraints and emphasize the importance of allowing students to express preferences while respecting feasibility constraints [Graves et al., 1993]. Subsequent work has examined both the theoretical properties and empirical performance of auction-based systems, often focusing on issues of incentive compatibility, efficiency, and fairness.

Schutzman provides a comprehensive comparative analysis of course registration auctions across multiple institutions, highlighting both their strengths and limitations. A key insight from this work is that while auction mechanisms are effective at capturing preference intensity, they often fail to induce truthful bidding [Schutzman, 2016]. In particular, FPSB formats, such as the one previously mentioned at Colorado College, encourage students to strategically shade bids based on

expected competition from other students. Even in mechanisms with second-price or clearing price rules, students must reason about future opportunities and budget constraints, thus complicating strategic decision-making.

Such literature also identifies several core design challenges. First, auction mechanisms must balance efficiency with equity. While allocating seats to the highest bidders may maximize total social welfare, it can disadvantage students with fewer accumulated points or less strategic sophistication. Second, many real-world systems struggle to accommodate abnormal relationships between courses, such as required prerequisite courses, which limits their ability to fully capture student preference profiles. Finally, the presence of artificial currencies introduces complex incentives as students may choose to save or spend points strategically across multiple periods, potentially distorting allocations.

2.3 The Major-Size Externality

Current research documents institutional characteristics that play a profound role in shaping bid amounts. At Colorado College, analysis of long standing auction practice revealed variation in bid amounts among departments, which does create cause for concern for allocative efficiency. Will bid values be several orders larger in the computer science major, rather than the Southeast Asian history major? Auction based bidding levels the playing field within departments, but creates inequities across interdisciplinary efforts.

Take Christine for example. Christine wants to study biology, a notoriously competitive and large major at the University of Virginia. Bid values in a SPSB auction for a particular biology lecture are guaranteed to be competitive, with such a large application pool. On the other hand, Alice, a music major, operates in a much smaller department with flexible enrollment offerings, and plentiful empty seats. An issue emerges when the two students both want to take Dracula, and Alice's surplus of bid value allows her to more easily win a seat. Should Christine be disadvantaged for being a biology major?

This *major-size externality* creates issues in our auction based system that aren't present in the current random serial dictatorship setup. To address this issue, we suggest several possible solutions. Normalization of bid amounts across departments could eliminate this inconsistency, but enrollment fluctuation, and its inherent complexity for students contradicts our design principle for tractability and simplicity. Similarly, long term adaptations of course supply could eliminate bid inefficiencies. If a department isn't a competitive bidding environment, it is likely this department could have excess capacity relative to student interest. This solution also faces issues, as course prestige factors into course value. Offering 20 Dracula courses at UVA wouldn't get filled, primarily because a core feature of course value often is its scarcity. For simplicity' sake now, we won't consider this issue, but it does warrant further investigation.

2.4 Why Auction-Based Enrollment?

Taken together, prior implementations and academic studies demonstrate that auction-based course allocation is both feasible and widely used, but there remains imperfections to be improved upon. Existing systems rely either on relatively simple formats that induce strategic inefficiencies (ex. first-price auctions) or more complex multi-round mechanisms that are difficult for students to understand and navigate accordingly.

In this work, we focus on the design and analysis of a simplified simultaneous SPSB framework for course allocation at the University of Virginia. By emphasizing simplicity and tractability, our approach aims to capture much of the efficiency gain from auction-based systems while reducing the burden on students to understand a complicated model's strengths and weaknesses. In doing

so, we build on prior work while exploring a design point that lies between highly expressive (but admittedly overly complex) combinatorial mechanisms and overly simplistic allocation rules.

3 Algorithm Design

In this section, we describe the implementation of our auction-based course allocation mechanism and the simulation framework used to evaluate its performance. Our goal is to design a tractable and interpretable system that closely mirrors the theoretical structure of a simultaneous sealed-bid auction while remaining simple enough for practical deployment.

3.1 Model Setup

We model course allocation as a resource allocation problem between a set of students and a set of courses. Let N denote the set of students and C denote the set of courses. Each course $j \in C$ has a fixed capacity k_j, representing the maximum number of students that can be enrolled.

Each student $i \in N$ is associated with a valuation vector $v_i = (v_{ij})_{j \in C}$, where v_{ij} represents the value that student i assigns to course j. These valuations capture heterogeneous preferences across students and courses. Additionally, each student is endowed with a fixed budget B_i, which constrains the total bids they may submit across all courses.

Students submit bids b_{ij} for each course j, subject to the budget constraint:

$$\sum_{j \in C} b_{ij} \leq B_i.$$

The outcome of the mechanism is an allocation matrix $x \in \{0, 1\}^{|N| \times |C|}$, where $x_{ij} = 1$ indicates that student i is assigned to course j.

3.2 Auction Mechanism

We implement a simultaneous sealed-bid auction in which all students submit bids for all courses at once. The mechanism proceeds in three main stages:

Bid Generation. For each student, bids are generated based on their underlying valuations. In our baseline implementation, bids are assumed to be truthful (i.e., proportional to v_{ij}), subject to the student's budget constraint. This allows us to evaluate the efficiency of the mechanism in an idealized setting without strategic distortion.

Allocation Rule. For each course j, students are ranked in descending order of their submitted bids b_{ij}. The course is then allocated to the top k_j bidders, ensuring that capacity constraints are satisfied. This process is conducted independently for each course, reflecting the simultaneous nature of the auction.

Pricing Rule. Allocated students pay a clearing price determined by the marginal losing bid for each course. Specifically, for course j, the price is set equal to the $(k_j + 1)$-th highest bid. This pricing rule approximates a second-price auction and is intended to reduce incentives for bid shading relative to first-price mechanisms.

This mechanism strikes a balance between simplicity and economic structure. While it does not capture complementarities between courses, it retains key desirable properties such as transparency and tractability.

3.3 Simulation Framework

To evaluate the performance of the proposed mechanism, we implement a simulation framework that generates synthetic course allocation environments. The simulation is parameterized by the number of students, the number of courses, and the number of independent runs used for aggregation.

Each simulation proceeds as follows:

(1) Generate a set of students and courses with randomly sampled valuations and capacities.
(2) Construct bid profiles for each student subject to their budget constraints.
(3) Execute the auction mechanism to produce an allocation and associated prices.
(4) Compute efficiency and pricing metrics for the resulting allocation.

The framework supports repeated runs with different random seeds, allowing us to analyze the distribution of outcomes and report aggregate statistics such as means and standard deviations. This provides a more robust evaluation of mechanism performance under varying conditions.

3.4 Evaluation Metrics

We evaluate the quality of the auction outcomes using several standard metrics from the mechanism design and resource allocation literature.

Total Welfare. The total welfare of an allocation is defined as:

$$W = \sum_{i \in N} \sum_{j \in C} v_{ij} x_{ij},$$

which measures the total value derived by all students from their assigned courses.

Optimal Welfare. We compute an estimate of the maximum achievable welfare W^* by solving the allocation problem optimally, subject to capacity constraints. This serves as a benchmark for evaluating the efficiency of the auction mechanism.

Efficiency Gap. The efficiency gap is defined as:

$$\text{Efficiency Gap} = \frac{W^* - W}{W^*},$$

which captures the relative loss in welfare compared to the optimal allocation.

Allocation Rate. The allocation rate measures the fraction of students who receive at least one course assignment.

Capacity Utilization. This metric measures the proportion of total course capacity that is filled under the allocation.

Revenue. Revenue is defined as the total payments collected from students based on the clearing prices of the courses to which they are assigned.

Pricing Metrics. We also examine the distribution of clearing prices, including the average price and the observed range across courses, to better understand how demand heterogeneity manifests in the auction outcomes.

Together, these metrics provide a comprehensive view of the performance of the mechanism, capturing both efficiency and practical considerations such as utilization and pricing behavior.

4 Performance Evaluation

Here, we evaluate the performance of the proposed auction mechanism using the simulation framework described in Section 3. We present results from two representative configurations to illustrate how the mechanism performs in both small and moderately scaled environments.

4.1 Experimental Setup

We consider two simulation settings. The first consists of a small market with 20 students and 5 courses, executed for a single run. The second scales the environment to 100 students and 10 courses, with results aggregated over 3 independent runs. In both cases, student valuations are generated randomly, and students submit bids subject to fixed budget constraints. Course capacities are fixed and binding.

4.2 Results

4.2.1 Small-Scale Environment. Figure 2 summarizes the results for the 20-student, 5-course configuration.

Metric	**Value**
Total Welfare	17,100.34
Optimal Welfare	17,314.75
Efficiency Gap	1.24%
Revenue	1,037.96
Allocation Rate	100.0%
Capacity Utilization	12.1%
Average Clearing Price	51.90
Price Range	38.16 – 80.33

Fig. 2. Simulation Results: 20 Students, 5 Courses (Single Run)

The results indicate that the auction achieves near-optimal efficiency, with only a 1.24% gap relative to the optimal allocation. All students receive at least one course assignment, demonstrating that the mechanism effectively distributes seats across participants. However, capacity utilization is relatively low at 12.1%, suggesting that total available seats significantly exceed demand in this configuration. This leads to relatively moderate clearing prices, with variation driven by differences in course demand.

4.2.2 Larger-Scale Environment. Figure 3 reports aggregated results for the larger configuration with 100 students and 10 courses, averaged over three simulation runs.

Metric	Mean	Std	Min	Max
Total Welfare	79,990.55	1,688.97	76,742.04	80,642.02
Optimal Welfare	80,532.16	1,628.63	78,236.62	81,842.79
Efficiency Gap	0.02	0.00	0.01	0.02
Revenue	6,393.38	686.90	5,745.51	7,344.17
Allocation Rate	1.00	0.00	1.00	1.00
Capacity Utilization	0.39	0.02	0.37	0.41
Avg. Price	63.93	6.87	57.46	73.44

Fig. 3. Simulation Results: 100 Students, 10 Courses (3 Runs)

In this larger setting, the mechanism continues to achieve high efficiency, with an average efficiency gap of approximately 2%. Total welfare remains close to the optimal benchmark across all runs, and the low standard deviation indicates stable performance across different realizations of the environment.

Capacity utilization increases significantly to approximately 39%, reflecting a more competitive environment in which seats are more fully allocated. Revenue also increases substantially, driven by higher aggregate demand and competition among students. The average clearing price rises to 63.93, with a wider observed range, indicating greater differentiation between high- and low-demand courses.

4.3 Discussion of Results

Across both configurations, the auction mechanism consistently produces allocations that are close to welfare-optimal. The efficiency gap remains small in both small and large environments, suggesting that the mechanism is robust to changes in scale.

A key observation is the relationship between competition and pricing. In the smaller environment, low capacity utilization results in relatively modest prices and limited competition. In contrast, the larger environment exhibits higher utilization and increased clearing prices, reflecting stronger competition for course seats. This demonstrates that the mechanism dynamically adjusts to demand conditions without requiring manual intervention or priority rules.

Additionally, the allocation rate remains at 100% across all experiments, indicating that every student is able to secure at least one course. This is an important practical consideration, as it suggests that the mechanism can provide broad access while still prioritizing high-value allocations.

4.3.1 Baseline Comparison. To contextualize these results, we compare the auction mechanism to a simplified baseline inspired by UVA's current enrollment system, modeled as a random serial dictatorship. Under this baseline, students are randomly ordered and sequentially assigned to their most preferred available course.

Unlike the auction mechanism, the baseline does not account for preference intensity and is sensitive to the ordering of students. As a result, high-value students may fail to obtain desired courses if they are assigned later in the sequence. This leads to lower expected welfare and greater variability in outcomes.

By contrast, the auction mechanism consistently achieves allocations that are closer to optimal, as demonstrated by the low efficiency gaps observed in Figures 2 and 3. These results highlight the advantage of incorporating bid-based preference intensity into the allocation process. To contextualize the performance of the auction mechanism, we compare it against a simplified baseline modeled as a *random serial dictatorship* (RSD) with seniority weighting, inspired by UVA's current course enrollment system. Under the RSD baseline, students are ordered pseudo-randomly subject to seniority priority, and each student is sequentially assigned to their most preferred available course. Critically, this mechanism is ordinal: it treats all preferences as binary and does not account for the intensity with which a student values one course over another.

This insensitivity to preference intensity is the core structural limitation of the RSD baseline. A student who marginally prefers course A over course B is treated identically to one who considers course A essential and course B nearly worthless. As a consequence, high-value students who draw unfavorable positions in the ordering may be systematically denied courses they value most, with no compensating mechanism to redirect scarce seats toward students whose welfare loss from non-allocation is greatest.

Our simulation results, summarized in Figure 4, confirm this theoretical limitation empirically. Across 20 runs with $n = 100$ students, $m = 4$ courses, and $r = 20$ seats per course, the auction mechanism yields a mean total welfare of \$80,015 (±\$4,712), compared to \$77,513 (±\$6,800) under RSD—a raw improvement of \$2,502 per simulation, or approximately 3.6%. Priority-weighted welfare, which discounts the welfare of students who already hold seniority advantages, similarly favors the auction: \$99,984 (±\$5,899) versus \$97,406 (±\$7,995), a difference of \$2,578 (2.9%).

Beyond the level gains, the auction mechanism also exhibits meaningfully lower outcome variance. The standard deviation of total welfare under RSD (\$6,800) exceeds that of the auction (\$4,712) by over 44%, reflecting the ordering-sensitivity of the serial dictatorship: students near the front of the queue are likely to be well-served regardless of preference intensity, while those at the back face high exposure to undesirable allocations. The auction, by contrast, allows students with strong

preferences to outbid competitors for contested courses, partially decoupling welfare outcomes from the arbitrary draw of the ordering.

Allocation rates are broadly similar across the two mechanisms—97.2% (±4.5%) for the auction and 97.7% (±4.6%) for RSD—indicating that neither mechanism produces substantial unmatched capacity under the simulated parameters. This near-parity in allocation rates underscores that the welfare advantage of the auction stems not from placing more students overall, but from placing them into courses they value more highly. The auction's near-perfect efficiency score of 99.9% (±0.4%) further confirms that the mechanism reliably identifies allocations close to the social optimum, even under competitive enrollment conditions.

Taken together, these results suggest that incorporating bid-based preference intensity into the allocation process yields consistent, measurable welfare improvements relative to ordinal priority mechanisms, without sacrificing seat utilization. Further equity metrics should be considered in further research.

Metric	**Auction**	**RSD**	**Δ (Auction − RSD)**
Total welfare	\$80,015 ± \$4,712	\$77,513 ± \$6,800	+\$2,502 (+3.6%)
Priority-weighted welfare	\$99,984 ± \$5,899	\$97,406 ± \$7,995	+\$2,578 (+2.9%)
Efficiency	99.9% ± 0.4%	—	—
Revenue	\$23,138 ± \$9,061	—	—
Allocation rate	97.2% ± 4.5%	97.7% ± 4.6%	−0.5%

Fig. 4. Simulation results: auction mechanism vs. random serial dictatorship (20 runs, n=100 students, m=4 courses, r=20 seats). Values reported as mean ± 1 SD.

5 Limitations

While the proposed auction-based mechanism demonstrates strong performance in simulation, several limitations should be noted. First, the model assumes independent valuations across courses and does not capture complementarities, such as prerequisite chains or course bundles that must be taken jointly. This restricts the expressiveness of student preferences and may lead to suboptimal allocations in cases where such relationships are important.

Second, our simulation assumes truthful bidding behavior, abstracting away from strategic considerations. In practice, students may engage in bid shading or attempt to anticipate competition, which could impact both efficiency and fairness. Although the second-price structure mitigates some incentives for manipulation, it does not fully eliminate strategic behavior.

Third, the results are based on synthetic data rather than real UVA enrollment data. As a result, the valuation distributions and demand patterns may not fully reflect real-world student preferences, particularly in high-demand or "bottleneck" courses.

Finally, the mechanism does not explicitly address fairness concerns across student groups, such as differences in major size or course availability. As discussed previously, these factors may introduce inequities that are not present in the current enrollment system and warrant further investigation.

These limitations highlight important directions for future work, including incorporating richer preference structures, modeling strategic behavior, and validating the mechanism using real-world data.

6 Conclusion

In this paper, we examined the problem of course enrollment at the University of Virginia as a resource allocation challenge under limited capacity and heterogeneous student preferences. We identified key inefficiencies in the current system, including the inability to capture preference intensity, the reliance on time-based priority, and the prevalence of strategic behavior under uncertainty.

To address these issues, we proposed a simultaneous sealed-bid auction framework in which students submit bids for individual courses subject to budget constraints. This mechanism provides a simple and tractable alternative to more complex combinatorial approaches while still allowing students to express relative preferences across courses. Through simulation, we demonstrated that the proposed mechanism consistently achieves allocations that are close to welfare-optimal, with low efficiency gaps and high allocation rates across both small and larger-scale environments.

Our results further show that the mechanism adapts naturally to differences in demand, with clearing prices reflecting competition for high-value courses and capacity utilization increasing in more competitive settings. Compared to a baseline random serial dictatorship model, the auction-based approach more effectively aligns course assignments with student valuations, highlighting the benefits of incorporating bid-based preference intensity into the allocation process.

While the model is intentionally simplified, it provides a promising foundation for rethinking course enrollment systems in practice. Future work may extend this framework to incorporate richer preference structures, address fairness concerns, and evaluate performance using real-world data. Overall, our findings suggest that auction-based enrollment mechanisms represent a viable and effective alternative to existing systems, offering meaningful improvements in both efficiency and flexibility.

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